



Interactive Animations and Games

Lesson 6

Random Numbers

Warm Up



Journal Prompt:

So far, our programs have done the same thing every time that we run them.

Are there any times that you'd want a program to do something differently each time it was run?

Question of the Day

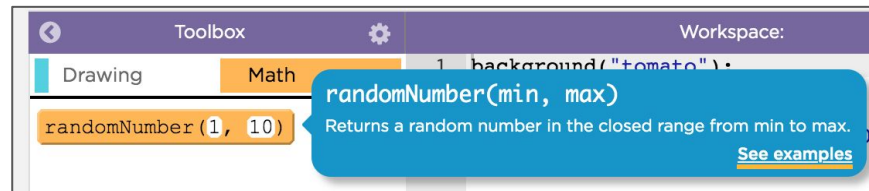
How can we make our programs behave differently each time they are run?

Activity



Code Studio, Level 1

Predict



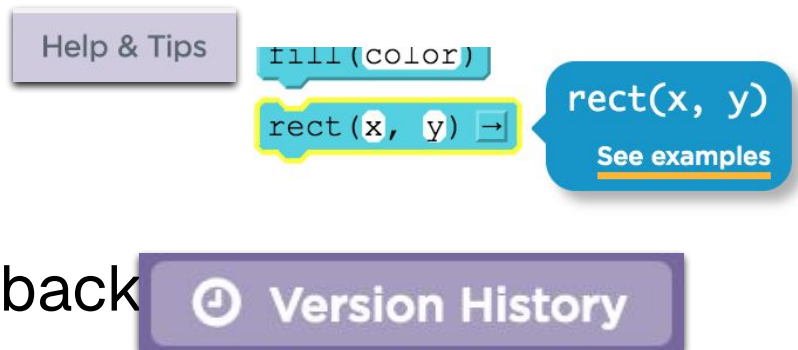
- The code introduces a new math block, `randomNumber()`.
- Try to predict what will happen when you press the run button.
- Write down what you think will happen when you press the run button
- Run the code
- Compare your screen with a neighbor's - did you both get the same thing?

Code Studio, Levels 2-5

Skill Building

Do This

- Follow the instructions on each level
- If you get stuck, try...
 - Help and Tips Tab
 - Code Documentation
- If you make a big mistake...
 - Use version history to go back

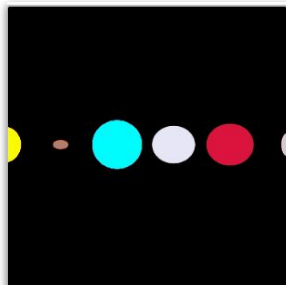


Code Studio, Level 6

Practice Level

Do This

- Practice your new skills on these levels
- Check with your teacher before moving on



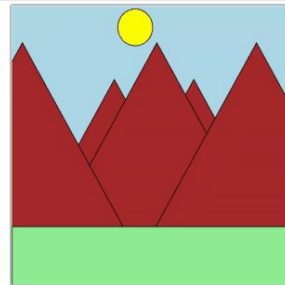
a Modify the Planets

Make the planets different sizes using random numbers.



b Debug: Random Shape Position

Keep the rectangles stacked on top of each other by changing the way random numbers are used.



c Modify the Scene

Randomize the location of items in the scene using random numbers.



Code Studio, Level 7

Assessment Level

Do This

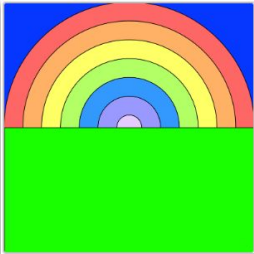
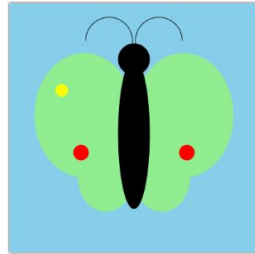
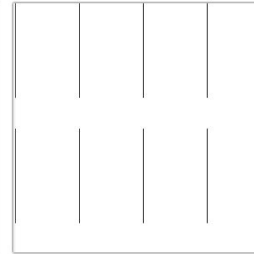
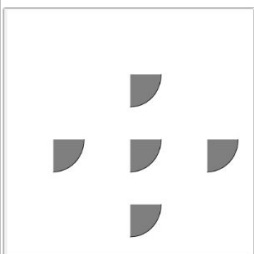
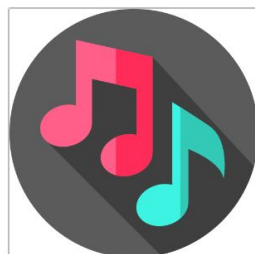
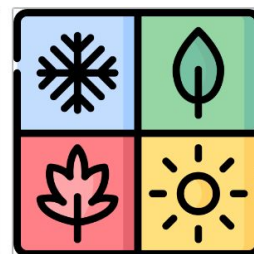
- Follow the instructions in Code Studio
- The "Rubric" tab describes how to demonstrate your understanding
- Check out the Help & Tips tab if you need help

Code Studio, Level 8

Challenge Level

Do This

- Challenge yourself on these levels.
- Learn new blocks or use the same ones in new ways.

 a Random RGB Randomize the colors of this rainbow.	 b Symmetry Use math to restrict the range of random numbers.	 c Random Lines Randomize all the properties of these lines.
 d Random Arcs Randomize all the properties of these arcs.	 e Favorite Song Use what you've learned so far to create a drawing that represents your favorite song!	 f Favorite Season Use all that you have learned so far to create a drawing of your favorite season that changes each time you run the program!

Wrap Up





Journal:

So far, we've only looked at random numbers.
Are there any other things that you might like
to be random in your program?



Question of the Day

How can we make our programs behave differently each time they are run?